Structured Programming

Part 2: Visualizing Algorithms with Flowcharts

By the end of this lesson, you should be able to:

- Explain how flowcharts represent algorithms visually.
- Identify the common symbols used in programming flowcharts.
- Represent sequence, selection, and iteration using flowcharts.

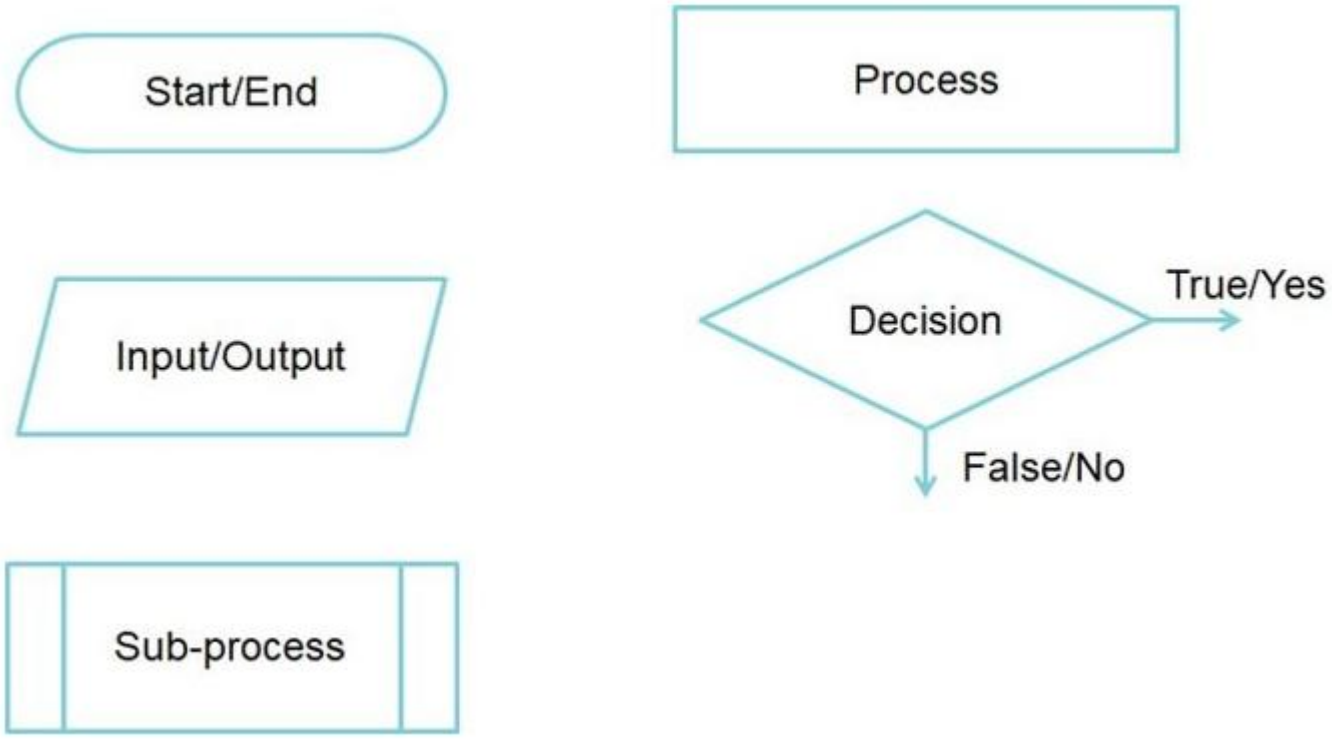
Flowcharts

A **flowchart** is a diagram that uses symbols and arrows to represent the steps and decisions in an algorithm.

Flowcharts allow programmers to plan and understand the logic of a program before writing code.

Arrows connect the symbols and show the order in which the steps occur.

Different symbols have different meanings.



I use this website to create flowcharts for free and without needing to make an account!

The link will also be shared on Schoology

https://app.diagrams.net/



START/END

The **START/END** symbol indicates where a process begins or ends.

Every flowchart should have at least one start symbol and one end symbol.



Process

The **Process** symbol represents an action or operation being performed.

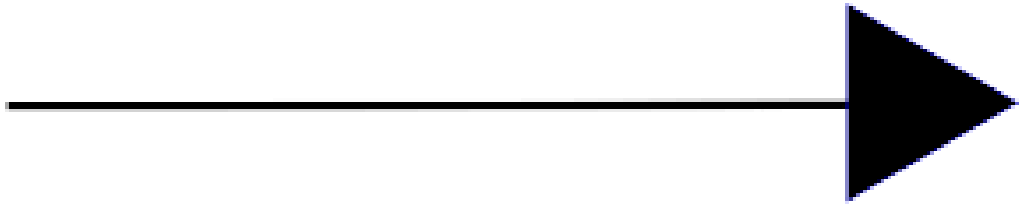
Most statements in a program are represented using process symbols.



Arrow

Arrows show the direction that control flows through the process.

They connect flowchart symbols and indicate what step happens next.



Input/Output

The **Input/Output** symbol represents data entering or leaving a program.

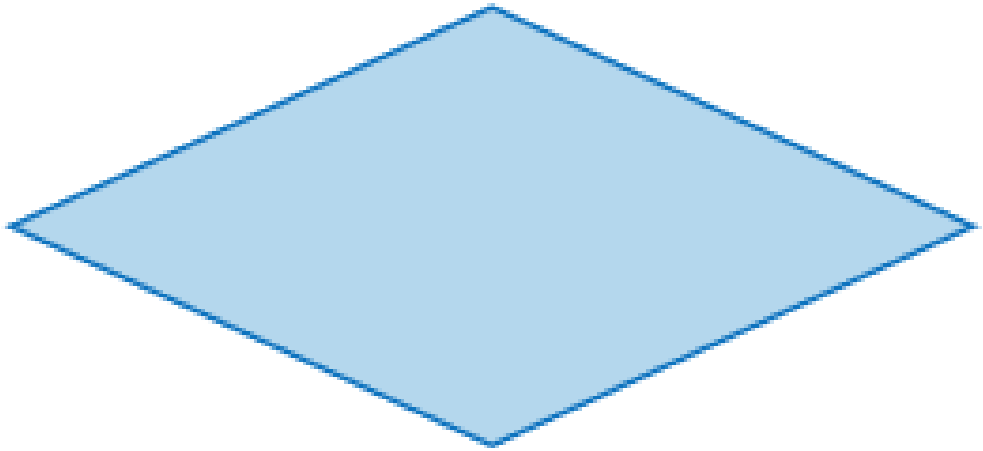
Common examples include receiving user input or displaying output to the screen.



Decision

The **Decision** symbol represents a condition that can evaluate to either true or false.

Depending on the result, the flow follows a different path.



Subprocess

The **Subprocess** symbol represents a task that has been defined elsewhere, such as a function.

It allows complex processes to be broken into smaller, reusable parts.

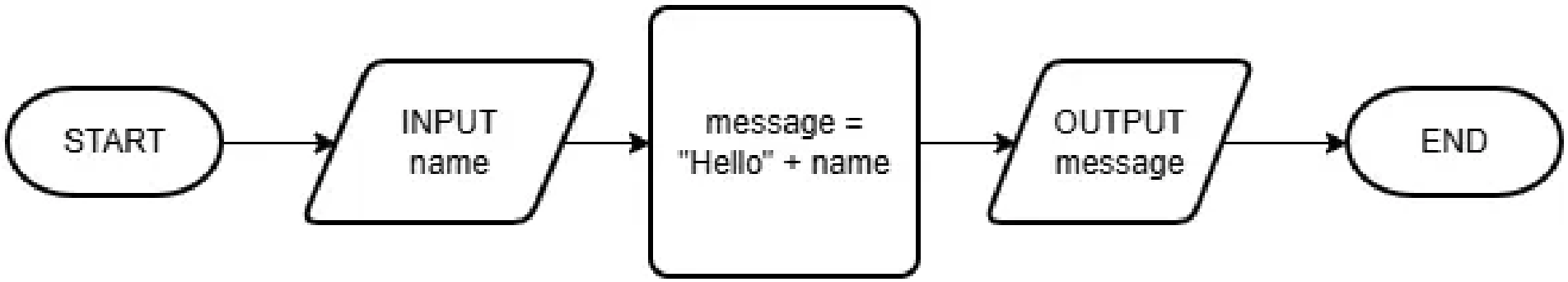


Visualizing Control Structures

Flowcharts can be used to visualize the three fundamental control structures used in programming: sequence, selection, and iteration.

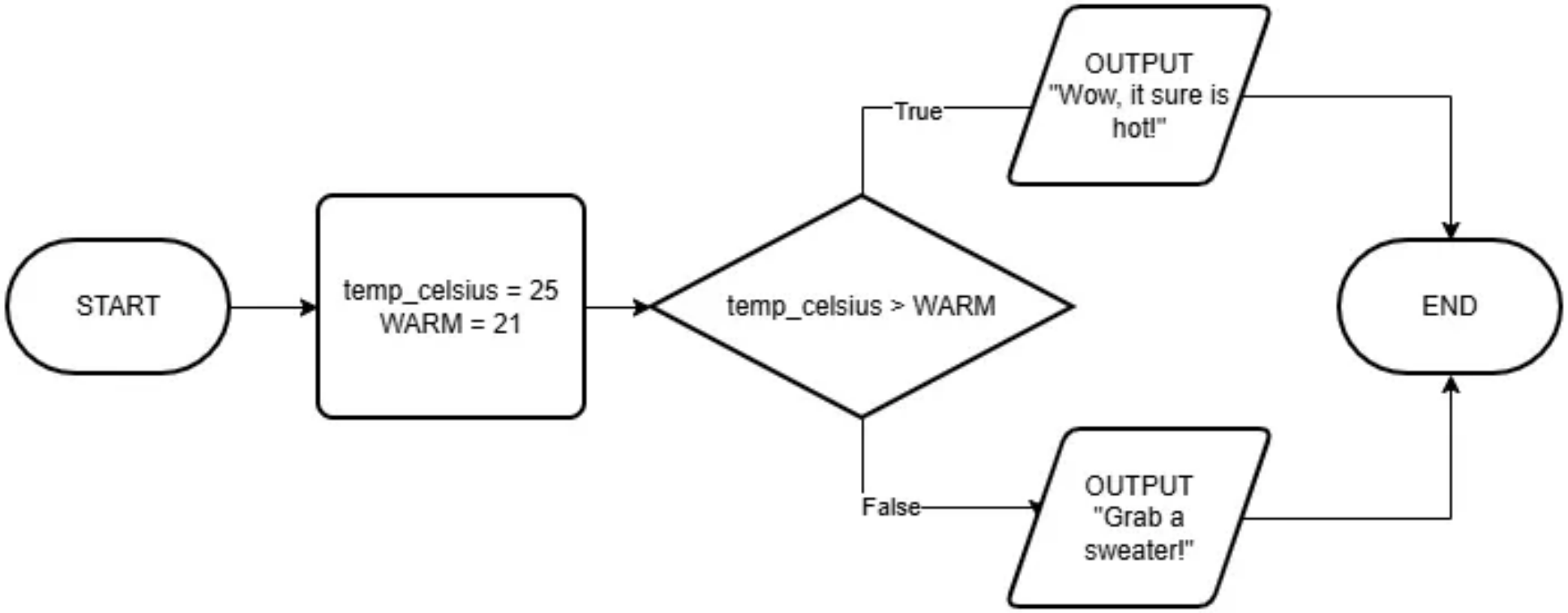
Visualizing Sequence

A sequence is represented by a series of process symbols connected by arrows.



Visualizing Selection

Selection structures use decision symbols to show different paths that may be taken based on a condition.



Visualizing Iteration

Iteration structures use arrows that loop back to an earlier point in the flowchart, representing code that repeats until a condition is met.

